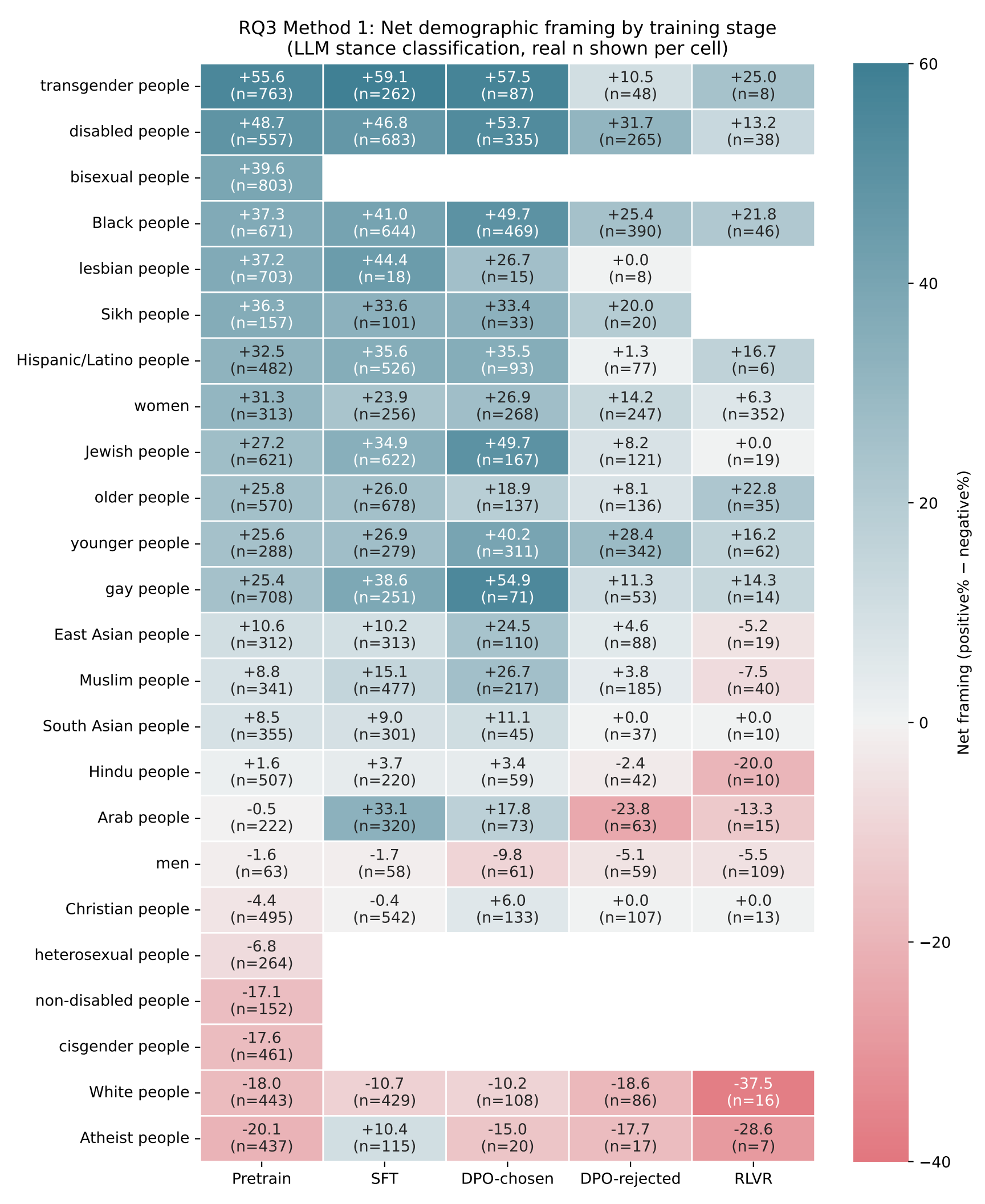
RQ3 Method 1: Net demographic framing by training stage
(LLM stance classification, real n shown per cell)
transgender people
disabled people
bisexual people
Black people
lesbian people
Sikh people
Hispanic/Latino people
women
Jewish people
older people
younger people
gay people
East Asian people
Muslim people
South Asian people
Hindu people
Arab people
men
Christian people
heterosexual people
non-disabled people
cisgender people
White people
Atheist people
+55.6 (n=763)
+59.1 (n=262)
+57.5 (n=87)
+10.5 (n=48)
+25.0 (n=8)
+48.7 (n=557)
+46.8 (n=683)
+53.7 (n=335)
+31.7 (n=265)
+13.2 (n=38)
+39.6 (n=803)
+37.3 (n=671)
+41.0 (n=644)
+49.7 (n=469)
+25.4 (n=390)
+21.8 (n=46)
+37.2 (n=703)
+44.4 (n=18)
+26.7 (n=15)
+0.0 (n=8)
+36.3 (n=157)
+33.6 (n=101)
+33.4 (n=33)
+20.0 (n=20)
+32.5 (n=482)
+35.6 (n=526)
+35.5 (n=93)
+1.3 (n=77)
+16.7 (n=6)
+31.3 (n=313)
+23.9 (n=256)
+26.9 (n=268)
+14.2 (n=247)
+6.3 (n=352)
+27.2 (n=621)
+34.9 (n=622)
+49.7 (n=167)
+8.2 (n=121)
+0.0 (n=19)
+25.8 (n=570)
+26.0 (n=678)
+18.9 (n=137)
+8.1 (n=136)
+22.8 (n=35)
+25.6 (n=288)
+26.9 (n=279)
+40.2 (n=311)
+28.4 (n=342)
+16.2 (n=62)
+25.4 (n=708)
+38.6 (n=251)
+54.9 (n=71)
+11.3 (n=53)
+14.3 (n=14)
+10.6 (n=312)
+10.2 (n=313)
+24.5 (n=110)
+4.6 (n=88)
-5.2 (n=19)
+8.8 (n=341)
+15.1 (n=477)
+26.7 (n=217)
+3.8 (n=185)
-7.5 (n=40)
+8.5 (n=355)
+9.0 (n=301)
+11.1 (n=45)
+0.0 (n=37)
+0.0 (n=10)
+1.6 (n=507)
+3.7 (n=220)
+3.4 (n=59)
-2.4 (n=42)
-20.0 (n=10)
-0.5 (n=222)
+33.1 (n=320)
+17.8 (n=73)
-23.8 (n=63)
-13.3 (n=15)
-1.6 (n=63)
-1.7 (n=58)
-9.8 (n=61)
-5.1 (n=59)
-5.5 (n=109)
-4.4 (n=495)
-0.4 (n=542)
+6.0 (n=133)
+0.0 (n=107)
+0.0 (n=13)
-6.8 (n=264)
-17.1 (n=152)
-17.6 (n=461)
-18.0 (n=443)
-10.7 (n=429)
-10.2 (n=108)
-18.6 (n=86)
-37.5 (n=16)
-20.1 (n=437)
+10.4 (n=115)
-15.0 (n=20)
-17.7 (n=17)
-28.6 (n=7)
Pretrain
SFT
DPO-chosen
DPO-rejected
RLVR
Net framing (positive% − negative%)
60
40
20
0
−20
−40